ZVS-Constrained Optimal Modulation LUT and Sampled Command-to-Current Gain of a Triple-Phase-Shift Dual Active Bridge

Contributions.

1. **Command-to-current gain of an optimal-modulation LUT.** An ideal inverse modulation delivers exactly the commanded current, so its gain from command to delivered current is unity. A stored, bilinearly interpolated LUT does not. Since the controller samples the current once per switching period, the gain it actually sees is the sampled gain g_s.

2. **g_s as a deployability metric for a fixed PI controller.** An interpolation error is a bias, which the integral action removes. A negative g_s inverts the sign of the feedback, and the controller then amplifies the error. The widest reversal along the charging ramp limits the usable crossover frequency, and the spread of g_s sets the margins the PI controller is tuned for.

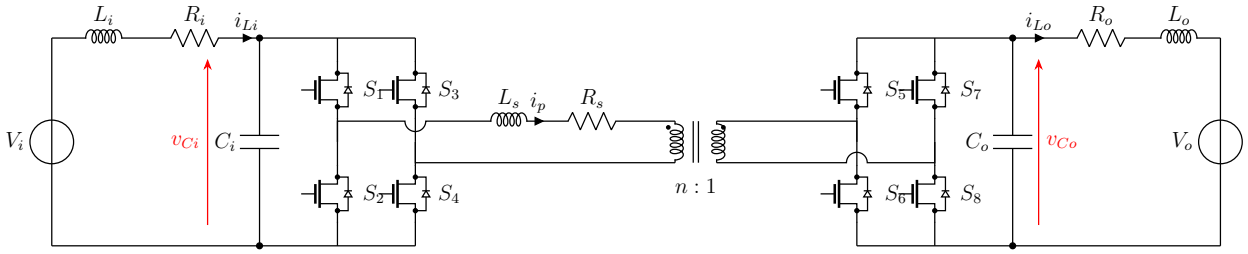


Figure 1: DAB circuit schematic with primary-referred AC-link inductance and resistance.

The duty variables D_1, D_2 and D_3 follow the edge-referenced TPS convention: $D_1, D_2 \in (0, 1]$ are the inner shifts of the primary and secondary bridges and D_3 is the displacement between the rising edges of the primary and secondary reference legs.

The modulation is computed on the ideal TPS waveform: the converter is lossless, the DC voltages are constant, the filters are not included, and the secondary is referred to the primary. The bridge voltages are $v_p \in \{-V_i, 0, V_i\}$ and $v_s \in \{-nV_2, 0, nV_2\}$, and between the eight sorted switching edges the link current is piecewise linear,

$$\frac{di_p}{dt} = \frac{v_p - v_s}{L_s}, \qquad \frac{1}{T_s}\int_0^{T_s} i_p\, dt = 0. \tag{1}$$

The transferred power $P(\mathbf{D})$, the delivered average current $\bar{i}_2(\mathbf{D}, V_2) = P(\mathbf{D})/V_2$ and the RMS current $I_{\mathrm{RMS}}(\mathbf{D})$ follow in closed form from (1).

At each switching edge k, the link inductance and the output capacitances of the commutating leg form a resonant circuit with

$$Z_p = \sqrt{\frac{L_s}{2C_{\text{oss}}}}, \qquad Z_s = n\, Z_p, \tag{2}$$

on the primary and secondary side, with a constant charge-equivalent $C_{\text{oss}} = 1$ nF. The commutation completes if the link current at the edge flows in the direction that charges the leg and its energy in L_s is large enough to swing the leg voltage across the bus. The ZVS margin z_k of edge k is the link current at the edge minus the current this requires. Dead time is not modelled.

At every operating point (I_{cmd}, V_2) the ZVS-constrained optimal modulation is

$$\begin{aligned} \mathbf{D}^\star(I_{\text{cmd}}, V_2) = \arg\min_{\mathbf{D}}\ & I_{\text{RMS}}(\mathbf{D}) \\ \text{s.t.}\ & \bar{i}_2(\mathbf{D}, V_2) = I_{\text{cmd}}, \\ & z_k(\mathbf{D}) \geq 0, \quad k = 1, \ldots, 8, \\ & D_1, D_2 \leq 1. \end{aligned} \tag{3}$$

It is stored in a LUT of 40×17 operating points, $I_{\text{cmd}} \in [5, 122.5]$ A and $V_2 \in [200, 1000]$ V, and interpolated bilinearly. The converter follows the rated charging profile

$$I_{\text{ref}}(V_2) = \min\left(122\ \text{A},\ \frac{P_{\text{rated}}}{V_2}\right), \qquad P_{\text{rated}} = 50\ \text{kW}. \tag{4}$$

Ideal gain. The equality constraint of (3) makes $\mathbf{D}^\star$ an inverse of the modulation,

$$\bar{i}_2\big(\mathbf{D}^\star(I_{\text{cmd}}, V_2), V_2\big) = I_{\text{cmd}}. \tag{5}$$

Where $\mathbf{D}^\star$ varies smoothly with I_{cmd}, that is, inside one operating mode, the chain rule applied to (5) at fixed V_2 gives

$$\bar{g}^\star = \frac{d\bar{i}_2}{dI_{\text{cmd}}} = \nabla_{\mathbf{D}}\bar{i}_2 \cdot \frac{\partial \mathbf{D}^\star}{\partial I_{\text{cmd}}} = 1. \tag{6}$$

The ideal command-to-current gain is unity.

Sampled gain. A stored, bilinearly interpolated LUT does not satisfy (5) between its stored points, so its gain is not unity. The controller does not see $\bar{i}_2$ either. It samples the delivered current $i_{2,s}$ once per switching period, at the start of the period. The gain it sees is evaluated at periodic steady state by a central difference with $h = 0.5$ A at fixed V_2,

$$g_s \approx \frac{i_{2,s}(I_{\text{cmd}} + h) - i_{2,s}(I_{\text{cmd}} - h)}{2h}. \tag{7}$$

What g_s gives. The sign of g_s separates a harmless error from a controller failure. A small interpolation error is a bias: the loop gain stays positive and the integral action of the PI

controller removes it. Where $g_s < 0$, a command correction moves the delivered current in the opposite direction, and a fixed-sign controller drives the current away from the reference.

The spread of g_s sets the robustness margins. The PI controller $C(z)$ of each voltage range is tuned on the command-to-sampled-current plant $G(z)$, including a one-period delay, scaled to the median positive g_s of that range, g_{50}. It is accepted when the loops scaled to the 10th and 90th percentiles,

$$L_p(z) = C(z)\,G(z)\,\frac{g_p}{g_{50}}, \qquad p \in \{10, 90\}, \tag{8}$$

keep PM $\geq 45^\circ$ and GM ≥ 6 dB.